

Multicategory Classification Based on the Hypercube Self-Organizing Mapping (SOM) Scheme

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Abstract. A new multiclass recognition strategy is proposed in this paper, where the self-organizing mapping (SOM) scheme with a hypercube mapped space is used to represent each category in a binary string format and a binary classifier is assigned to each bit in the string. Our strategy outperforms the existing approaches in the prior knowledge requirement, the number of binary classifiers, computation complexity, storage requirement, decision boundary complexity and recognition rate.

1 Introduction

In a multicategory classification problem, if each adjacent class pair can be robustly separated with a binary classifier, all of the κ classes can be consequentially separated by the fusion of all binary classifiers. The result is that a binary code representation for each category, which is of a string format, is obtained, and each bit in the string corresponds to a binary classifier. In this approach, the optimal hypercube mapped space, which can preserve topological structure of the multiclass dataset and separation information among separate classes as largely as possible, is the key for getting simple decision boundaries and a small number of classifiers. The former, implemented with a self-organizing mapping (SOM) scheme [1] due to its topological order preservation property, and the latter, implemented with the hypercube structured mapped space in the SOM and our proposed circulation training algorithm (CiTA) for training the SOM due to its good separation structure, are used simultaneously and their tradeoff is made for solving a multicategory classification problem with smaller number of binary classifiers and each classifier corresponding to a simpler decision boundary.

2 Hypercube SOM Scheme Based Multicategory Classification

2.1 Ideal Cases

Following assumptions on training data are the ideal case for the start of our discussions: a) compactness and exclusiveness of the classes in the training data, i.e., samples belonging to each class form a compact cluster, and each compact cluster includes samples belonging to only this class; b) exclusive center-vertex mapping, i.e., the κ class centers can be exclusively mapped to κ vertices of a hypercube of the

smallest dimensions by the hypercube SOM, of which separate class centers can be mapped to separate vertices.

If the above two assumptions hold simultaneously for the training data, at the convergence of the learning process of the SOM trained with the centers of classes in the data, an exclusive mapping from the centers of classes in the input space to the vertices of the M -dimensional hypercube ($M = \lceil \log_2 K \rceil$) in the mapped space is obtained. We denote each vertex in the hypercube as its coordinate which is a string of M -bit length, $(z_1 z_2 \cdots z_j \cdots z_M)$ with $z_j \in \{0,1\}$ for $j=1,2,\dots,M$.

Notice that the hamming distance between the vertices coded with $(z_1 z_2 \cdots z_{j-1} 0 z_{j+1} \cdots z_M)$ (i.e. $z_j = 0$) and with $(z_1 z_2 \cdots z_{j-1} 1 z_{j+1} \cdots z_M)$ (i.e. $z_j = 1$) is one, which is the smallest hamming distance for different vertices in the hypercube, no matter what values of $z_1, z_2, \dots, z_{j-1}, z_{j+1}, \dots, z_M$ are. Therefore, from the topological order preservation of the hypercube SOM mapping from input space to hypercube vertices, the class samples belonging to the class code of $(z_1 z_2 \cdots z_{j-1} 0 z_{j+1} \cdots z_M)$, denoted by x_{j0} , and the class samples belonging to the class code of $(z_1 z_2 \cdots z_{j-1} 1 z_{j+1} \cdots z_M)$, denoted by x_{j1} , has the smallest topological distance in the input space, meaning that they are adjacent in the input space, and should be separated by a decision boundary. Therefore, the j^{th} binary classifier should be trained with the training data $\Omega_j = \{(x_i, z_j^{(i)}) \mid z_j^{(i)} = 0 \text{ for } x_i \in X_{j0}; z_j^{(i)} = 1 \text{ for } x_i \in X_{j1}\}$ for $j=1,2,\dots,M$. This means that each bit of the class code corresponds to a binary classifier, and altogether M binary classifiers, each trained with all the N training samples, are needed for the K -class recognition problem. We take an example of $K=6$ and $M = \lceil \log_2 K \rceil = 3$ for showing the mechanism of the hypercube SOM scheme for solving the K -class recognition problem.

Two test calibrations are studied for multicategory classification based on hypercube SOM scheme. In the first test calibration, referred to as the sign test calibration, a test sample x is inputted to the j^{th} binary classifier to get its corresponding belonging $z_j \in \{0,1\}$ for $j=1,2,\dots,M$. $(z_1 z_2 \cdots z_j \cdots z_M)$ is then obtained when x is inputted to all the M binary classifiers. The class label y of the test sample x is directly the one which corresponds to the class code $(z_1 z_2 \cdots z_j \cdots z_M)$ in the list form of the correspondences between class labels and class codes. Another test calibration is what we call the nearest neighbor test calibration. The test sample x is inputted to the j^{th} binary classifier for getting its corresponding discriminatory function value $z_j \in [0,1]$ for $j=1,2,\dots,M$. Then we let the test sample x belong to the class label of its closest non-vacant vertex in the hypercube.

2.2 Some Problems in Realistic Cases

The problems which embarrass its realistic applications and need to be tackled includes: a) the training data are compact and exclusive but could not mapped onto the vertices of the smallest-scale hypercube SOM; b) the training data are not compact or exclusive.

For problem b), due to the clustering distribution of each class, class center is meaningless, but cluster center is meaningful. However, the clustering structure of the

data is unknown in prior, which includes how many clusters there is for each class of data, and how samples belonging to that class of data is clustered in the input space. Any clustering algorithm which can find both the solutions can be employed. In our experiments mentioned in this paper, we select the rival penalized competitive learning (RPCL) algorithm [2] as the clustering algorithm. Generally speaking, cluster centers of different classes in the data may not always be mapped onto exclusive vertices of the hypercube. Therefore, we should pursue two directions for modifying the above method used in the ideal cases to be suitable to more realistic applications: a) to enlarge the scale of the hypercube, such that it becomes smallest without overlapped vertices in the hypercube SOM; b) to force the cluster centers, which are mapped to overlapped vertices, to be moved to their most adjacent vacant/same-class vertices. Due the unawareness of the clustering structure of the data in prior, we suggest to have a tradeoff for getting simple decision boundaries and a small number of classifiers. Hence, an exclusively mapped class ratio (EMCR) is introduced, which is defined as the ratio between the number of the exclusively mapped classes to that of the all classes, to be compared with a predetermined threshold for determining whether the hypercube mapped space should be enlarged or overlapped cluster centers should be moved. The whole procedure referred to as CiTA for training SOM is as follows.

Parameter: EMCR threshold t_h .

Input:

an initial hypercube structure as the mapped space; a set of cluster centers, and their belongings to classes.

Output:

a hypercube SOM of a suitable size.

learning process:

Step 1: learn the map with cluster centers of all the classes with conventional SOM learning algorithm; calculate the EMCR from the obtained map;

Step 2: repeat this step until the EMCR equals one:

If the $EMCR > t_h$, prohibit the update of synaptic weight vectors relating to exclusive vertices, set the obtained weight vectors relating to other vertices to be their initial ones and keep the mapped space unchanged; continue to train the map with only cluster centers which are mapped onto mixed vertices with conventional SOM learning algorithm; calculate the EMCR from the obtained map;

Otherwise, enlarge the hypercube, and go to Step 1.

3 Experimental Results

The digit dataset consisting of handwritten numerals (“0”~“9”) was extracted from a collection of Dutch utility maps. Details of this dataset are available in the literature [3]. In our experiments we always used the same subset of 1000 Karhunen-Loève feature samples for testing and various subsets of the remaining 1000 patterns for training. After the RPCL, there are 2, 2, 3, 2, 3, 3, 3, 2 and 2 cluster centers for the classes “0, 1, 2, 3, 4, 5, 6, 7, 8 and 9” respectively. Nevertheless, by our $2 \times 2 \times 2 \times 2$ structured hypercube SOM based multicategory classification strategy, the cluster centers of each class are mapped onto one vertex of the hypercube in the class coding phase

Table 1. The recognition results of the OAA based SVM strategy and the hypercube SOM based strategy using binary SVMs for a set of handwritten digits

OAA based SVM strategy			
number of classifiers		10	
kernel		gaussian	poly
kernel parameter		9	3
number of support vectors		3136	1550
recognition rates (%)	sign	86.40	91.90
	maximum	97.70	97.40
hypercube SOM based strategy using binary SVMs			
hypercube		$2 \times 2 \times 2$	
number of classifiers		4	
kernel		gaussian	poly
kernel parameter		9	3
number of support vectors		1476	929
recognition rates (%)	sign	97.00	96.40
	nearest neighbor	98.10	97.50

with $EMCR_{Th} = 80\%$, and there are 6 vacant vertices. As shown in Table 1, our hypercube SOM based strategy for multicategory classification not only reduces the number of binary classifiers and the storage requirement, but also improves the recognition rate to some extent.

4 Conclusion

In this paper, an effective κ -class recognition strategy using a least number of binary classifiers with simplest decision boundaries is proposed. Recognition experiments show that the proposed method outperforms OAA based SVM classifier in the number of binary classifiers, computation complexity, storage requirement, decision boundary complexity and recognition rate.

References

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